Data Science 322: Final Project

Algorithmic Trading Bot with Alpaca

Instructor: P.M.H

Spring 2025

Summary

In this final project, student teams will design, implement, and backtest a day trading bot using the Alpaca API. However, before accessing the Alpaca environment, every team must demonstrate proficiency in financial data manipulation using Yahoo Finance and Python. This project tests your data science skills, algorithmic thinking, and ability to work effectively as a team.

Goals

The goals of this assignment are:

1. Demonstrate mastery of financial data retrieval and manipulation using Python (‘yfinance‘).
2. Develop and implement a programmatic trading strategy.
3. Interface with the Alpaca API for paper trading execution.
4. Foster professional team dynamics through reporting and team-building activities.

Phase 1: Instructor Check-off (Prerequisite)

Before starting the Alpaca bot, your team must be **checked off** by the instructor. You must demonstrate that you can effectively pull, clean, and visualize financial data using Yahoo Finance.

Getting Started with Yahoo Finance (‘yfinance‘)

Yahoo Finance is a popular source for stock market data. The ‘yfinance‘ library offers a threaded and Pythonic way to download market data from Yahoo! Finance.

Basic Usage Example

Below is a simple script to get you started. It installs the library, fetches data for Apple (AAPL), and prints the first few rows.

```python
# First, ensure you have the library installed:
# pip install yfinance

import yfinance as yf
import pandas as pd
import matplotlib.pyplot as plt
```

```python

# 1. Define the Ticker Symbol
ticker_symbol = "AAPL"

# 2. Get Data on this Ticker
ticker_data = yf.Ticker(ticker_symbol)

# 3. Get the Historical Prices for this Ticker
# Period options: 1d, 5d, 1mo, 3mo, 6mo, 1y, 2y, 5y, 10y, ytd, max
ticker_df = ticker_data.history(period='1d', start='2024-01-01', end='2025-01-01')

# The resulting DataFrame 'ticker_df' has columns:
# Open, High, Low, Close, Volume, Dividends, Stock Splits

print("Head of Data:")
print(ticker_df.head())

# 4. Simple Visualization
ticker_df['Close'].plot(title="AAPL Closing Price")
plt.show()
```

Listing 1: Basic Yahoo Finance Data Retrieval

Explanation of Concepts

- **Ticker Object:** The ‘yf.Ticker("SYMBOL")‘ creates an object that allows you to access various data points (financials, earnings, history) for that specific company.
- **History:** The ‘.history()‘ method is the workhorse of this library. It returns a Pandas DataFrame, which is the standard format for data science analysis.
- **Pandas Integration:** Because the data is returned as a DataFrame, you can immediately use standard pandas methods (like ‘.rolling()‘, ‘.mean()‘, or slicing) to calculate technical indicators (e.g., Simple Moving Averages).

Recommended Reading

To prepare for the check-off and the final project, read the following resources:

- **Documentation:** yfinance PyPI Official Documentation
- **Tutorial:** Stock Price Prediction Basics (GeeksforGeeks)
- **Concepts:** Basics of Algorithmic Trading (Investopedia)

Phase 2: Time Series Analysis Application

Once you have the data, you must apply the Time Series techniques learned in class to generate trading signals. A raw price chart is insufficient for a trading bot; the bot needs mathematical indicators to make decisions.

Mathematical Concepts

Your bot should rely on statistical indicators. A common starting point is the **Moving Average Crossover** strategy, which utilizes time-series smoothing.

Simple Moving Average (SMA)

The Simple Moving Average treats the price series as a sliding window. For a window of size k at time t, the SMA is defined as:

$$SMA_t^{(k)} = \frac{1}{k}\sum_{i=0}^{k-1} P_{t-i}$$

Where P_t is the closing price at time t.

Exponential Moving Average (EMA)

The EMA places greater weight on recent data points, making it more responsive to new information (conceptually similar to AR models). The recursive formula is:

$$EMA_t = \alpha \cdot P_t + (1-\alpha) \cdot EMA_{t-1}$$

Where α is the smoothing factor, typically calculated as $\alpha = \frac{2}{k+1}$.

Implementation Example

Below is an example of how to calculate these metrics and generate a "Signal" for your bot.

```python
# Calculate the 20-day (Fast) and 50-day (Slow) Moving Averages
ticker_df['SMA_20'] = ticker_df['Close'].rolling(window=20).mean()
ticker_df['SMA_50'] = ticker_df['Close'].rolling(window=50).mean()

# Define the Strategy:
# Signal = 1 (Buy) if SMA_20 > SMA_50
# Signal = 0 (Hold/Sell) if SMA_20 < SMA_50
ticker_df['Signal'] = 0
ticker_df.loc[ticker_df['SMA_20'] > ticker_df['SMA_50'], 'Signal'] = 1

# Calculate Daily Returns of the stock
ticker_df['Daily_Return'] = ticker_df['Close'].pct_change()

# Calculate Strategy Returns (Shift signals by 1 to avoid look-ahead bias)
ticker_df['Strategy_Return'] = ticker_df['Signal'].shift(1) * ticker_df['Daily_Return']

# Plot the strategy performance
(ticker_df['Strategy_Return'] + 1).cumprod().plot(title="Cumulative Strategy Returns")
```

Listing 2: Generating Signals with Time Series Analysis

Advanced Option: Students aiming for an 'A' are encouraged to explore more advanced Time Series models such as **ARIMA** (AutoRegressive Integrated Moving Average) or **GARCH** (volatility modeling) to predict future price movements rather than just following trends.

Phase 3: The Day Trading Bot (Alpaca)

Once checked off, you will proceed to the main project. You will use the **Alpaca Paper Trading API** to implement your strategy.

- You must not use real money. Use the Alpaca "Paper Trading" environment.
- Your bot should be capable of making buy/sell decisions based on live or historical data analysis.

- Further technical details on Alpaca keys and API endpoints will be provided after the Phase 1 check-off.

Team Reports and Team-Building Activities

Note: This component is designed to simulate a real-world data science team environment while keeping the experience engaging and enjoyable.

Weekly Team Reports

Each week, the instructor will briefly meet with the designated team lead to discuss the team's progress, division of responsibilities, and overall team dynamics. These informal check-ins focus on project status, collaboration, and any challenges the team may be facing.

Governance and Personnel Management (The "Real World" Simulation)

To mimic professional workplace accountability, teams will follow these rules every two weeks:

- **Leadership Rotation:** Every two weeks, the team must hold a vote to either retain the current team lead or elect a new one.
- **Termination Policy:** If a team member is not contributing or fails to assist the team, the team may vote to "fire" that individual.
- **The Job Market:** A fired student must use their resume to "interview" with other existing teams to find a new position.
- **Startup Teams:** If two or more students are teamless, they may form a new "startup" team.
- **Solo Mandate:** If no team is willing to hire a student due to poor work ethics or lack of performance, that student is required to complete the remainder of the Final Project solo.

Team-Building Activities

Teams will periodically participate in team-building activities led by the team lead with the goal of boosting team morale and strengthening collaboration.

- **Activities:** Activities may include (but are not limited to) group dinners, game nights, social outings, hikes, or similar events.
- **Documentation:** Teams will document participation by submitting a photo of the group together.
- **Purpose:** Creativity and enthusiasm are encouraged; this portion of the course is meant to promote strong teamwork, communication, and a positive working environment.

Final Project Report

Overview: This report should follow the basic structure of a scientific or technical paper. You are ***required*** to use `Overleaf/LaTeX` for writing your paper.

The basic parts of the final paper are as follows:

1. **Title and Author Information**

2. **Abstract:** A brief summary of your trading strategy and results.
3. **Introduction:** Motivation for the chosen strategy.
4. **Methodology & Code Explanation:**
   - Explanation of the ‘yfinance‘ data analysis.
   - Explanation of the Alpaca integration.
   - Logic behind the trading algorithm.
5. **Results:**
   - Performance metrics (ROI, Sharpe Ratio, etc.).
   - Backtesting results vs. Paper Trading results.
6. **Discussion:**
   - ***Note:*** If you encountered issues (API limits, latency, bugs), describe how you addressed them.
   - Reflect on the debugging process.
   - **Risk Analysis:** Discuss the limitations of your bot and potential risks if used with real capital.
7. **Conclusion**
8. **References**
9. **Appendices:**
   - ***Note:*** Include the full source code in the appendices.
   - **Team Photos:** Include the photos from your Team-Building Activities here.

Academic Integrity Policy

- **Cheating is strictly prohibited.** Copying directly from AI-generated content or any other published work without significant editing or proper citation will not be tolerated.
- Any instance of plagiarism or academic dishonesty will result in a score of zero for the assignment, and a report will be made to the school administration.
- Students are expected to submit their own original work.

Grading Rubric

Component	Expectations	Weight
Code Quality	Functional Python code for Yahoo Finance and Alpaca. Proper error handling, modular design, and comments. **Attach code to Appendix.**	30%
Results	Comprehensive evaluation of the bot's performance using ROI, Sharpe Ratio, or backtesting metrics. Evidence of critical thinking in analyzing why the strategy succeeded or failed.	30%
Results & Discussion	Analysis of the trading strategy performance. Critical reflection on limitations and risks.	20%
Writing Style	Well-organized, clear writing with proper grammar. Adherence to Overleaf format.	20%

Upload to Canvas the following:

- **PDF report:** `TeamName_FinalProject_DS322.pdf`
- **LaTeX code:** `TeamName_FinalProject_DS322.tex`
- **Code:** `TeamName_SourceCode.zip`